BOYU (BARRY) LIU

barryliuapp@gmail.com | 415-845-3878
ORCID: 0009-0004-8056-3141

Education

University of California, San Diego | La Jolla, CA 09/2021 - 06/2024
B.S., Molecular and Cell Biology | GPA: 3.77/4.00; Provost Honors, awarded 7 of 9 quarters.

Research Experience

Research Assistant (Full-Time) - Kosuri Lab, Salk Institute of Biological Studies 08/2024 - Present

- Currently leading a project using the tracking of RNA polymerase transcription as a potential new sequencing method. At this stage, I'm focused on registering the observed angular motion with nucleotide position on the template DNA.
- Assisted with the dcORBIT project with various experiments and troubleshooting. Designed and executed follow-up experiments to answer reviewer questions.
- Optimized purification of E.Coli RNA Polymerase Transcription elongation complexes via streptavidin magnetic beads 3 fold.
- Worked with the DNAntenna to: observe target-specific Cas9-mediated cleavage, observe Brownian motion and measure intrinsic physical properties of dsDNA with different lengths, measure and optimize ligation of different DNA origami structures.
- Mentored multiple lab members in technical knowledge and experimental design. Helped design rotation projects for two PhD students, one of whom has since joined the lab.

Undergraduate Intern (Part-Time) - Kosuri Lab, Salk Institute of Biological Studies 06/2023 - 08/2024

- Worked with various DNA origami structures: modifications, simulations of Brownian motion in OxDNA, and conducted high-speed imaging to observe Brownian motion.
- Designed an assay to measure individual strand ligation efficiency in a dsDNA overhang-mediated ligation reaction.

Research internship | Science Internship Program, UC Santa Cruz 06/2020 - 08/2020

- Ten-week summer bioinformatics research internship working with Dr. Rita Mehta's team on project Comparing Convergence, Biases, and Data Gaps in Carnivorous Marine Mammal Feeding Strategies.

Publications

Dye-cycling DNA origami rotors for long-term tracking of transcription at base-pair resolution
Wacker AL, Fantasia RJ, Tenner B, **Liu B**, Wu JD, Monell N, Kosuri P.
Cell Reports Methods (2026); 6: 101550. DOI: 10.1016/j.crmeth.2026.101550

Preprints

NanoNERF: A nanoscale NERF blaster replica made of DNA
Takiguchi LP, Rober MB, Olvera JG, Wacker AL, Fantasia RJ, **Liu B**, Shipley W, Tao A, Kosuri P.
bioRxiv (2023) [Preprint]. DOI: 10.1101/2023.10.02.560388

Presentations

Base-pair registration of ORBIT traces using U-tract motifs
McCool J, **Liu B**, Monell N, Wacker AL, Fantasia RJ, Coles AM, Kosuri P.
Poster, Salk Science at the Seaside Symposium, La Jolla, CA, October 2026.

NEON: Rotor tracking for scalable sequence discrimination via kinetic signatures
Monell N, **Liu B**, McCool J, Fantasia RJ, Wacker AL, Castiglione J, Kosuri P.
Poster, Salk Science at the Seaside Symposium, La Jolla, CA, October 2026.

Pause-site registration aligns ORBIT single-molecule trajectories for sequence-resolved transcription
Fantasia RJ, **Liu B**, Castiglione J, Monell N, Wacker AL, Tenner B, Kosuri P.
Poster, Biophysical Society 70th Annual Meeting, San Francisco, CA, February 2026.

Making DNA bending visible: A new tool to study Cas9
Liu B, Takiguchi LP, Tenner B, Fantasia RJ, Wacker AL, Shen K, Monell N, Kosuri P.
Poster, Salk Science at the Seaside Symposium, La Jolla, CA, October 2025.

DNAntenna: A modular DNA origami platform for dynamic tracking of DNA bending
Liu B, Takiguchi LP, Tenner B, Fantasia RJ, Wacker AL, Shen K, Monell N, Kosuri P.
Poster, Biophysical Society 69th Annual Meeting, Los Angeles, CA, February 2025.

DNAntenna: A modular DNA origami platform for tracking DNA bending
Takiguchi LP, Fantasia RJ, Wacker AL, **Liu B**, Shen K, Kosuri P.
Platform talk, Biophysical Society 68th Annual Meeting, Philadelphia, PA, February 2024.
Abstract: *Biophys J*. 2024;123(3 Suppl 1):173a.

Skills

Laboratory

- Nucleic acid nanotechnology: DNA origami design, folding, modification; PCR design and optimization; CRISPR-Cas9 design and *in vitro* cleavage; *in vitro* sgRNA synthesis.
- Proteins and Enzymes: E. coli RNA polymerase; SpyCas9, Spy-dCas9; T4 Ligase; restriction enzymes EcoRI, XhoI; proteinase K; various nucleases; GODCAT oxygen scavenging system.
- Labeling techniques: Digoxigenin/anti-Dig; biotin-streptavidin; SNAP tag labeling; NHS-ester labeling.
- Microscopy techniques: single-molecule TIRF microscopy; wide-field epifluorescent microscopy; assembling and imaging of custom tubing-based fluidics slides.
- General: streptavidin and poly-T magnetic bead purifications; polyacrylamide and agarose gel electrophoresis; bacterial and yeast culturing (courses).

Digital

- Scadnano, OxDNA, Oxview for DNA origami; coding in Python and HTML; AI-assisted scientific data analysis; Fiji/ImageJ; SnapGene; ChimeraX.

Other Skills

- Bilingual and biliterate in Mandarin and English.
- Extremely proficient as a bicycle mechanic.